LIMITS AND COLIMITS OF ABELIAN SHEAVES ON A SITE, AND EXACTNESS OF FILTERED COLIMITS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites four external items (two lemmas and one categories lemma, plus the presheaf lemma), all recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Limits and colimits of abelian presheaves are computed sectionwise. Stacks Project, Modules on Sites, Lemma lemma-limits-colimits-abelian-presheaves (tag 09B2). Supplies the presheaf-level descriptions that the sheaf-level statements are compared with.
- The limit of a diagram of sheaves of sets is a sheaf; left adjoints preserve colimits. Stacks Project, Sites, Lemma sites-lemma-limit-sheaf (tag 00W1); Categories, Lemma categories-lemma-adjoint-exact (tag 0036). Justify that the presheaf limit is a sheaf and that sheafification computes the colimit, hence exactness for filtered colimits.

Source reference. Original lemma 03CL: sources/stacks-chow/tags.txt, source lines 4388–4388. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00W2: sources/stacks-chow/tags.txt, source lines 1168–1168. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0038: sources/stacks-chow/tags.txt, source lines 125–125. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 05AA: sources/stacks-chow/tags.txt, source lines 6843–6843. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{C}$ be a site.*

(1) *All limits and colimits exist in $Ab(\mathcal{C})$.*
(2) *Limits are the same as the corresponding limits of abelian presheaves over $\mathcal{C}$ (i.e., commute with taking sections over objects of $\mathcal{C}$).*
(3) *Finite direct sums are the same as the corresponding finite direct sums in the category of abelian pre-sheaves over $\mathcal{C}$.*
(4) *A colimit is the sheafification of the corresponding colimit in the category of abelian presheaves.*
(5) *Filtered colimits are exact.*

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Proof. By Lemma 03CL limits and colimits of abelian presheaves exist, and are described by taking limits and colimits on the level of sections over objects.

Let $\mathcal{I} \to Ab(\mathcal{C})$, $i \mapsto \mathcal{F}_i$ be a diagram. Let $\lim_i \mathcal{F}_i$ be the limit of the diagram as an abelian presheaf. By Sites, Lemma 00W2 this is an abelian sheaf. Then it is quite easy to see that $\lim_i \mathcal{F}_i$ is the limit of the diagram in $Ab(\mathcal{C})$. This proves limits exist and (2) holds.

By Categories, Lemma 0038, and because sheafification is left adjoint to the inclusion functor we see that $\operatorname{colim}_i \mathcal{F}$ exists and is the sheafification of the colimit in $PAb(\mathcal{C})$. This proves colimits exist and (4) holds.

Finite direct sums are the same thing as finite products in any abelian category. Hence (3) follows from (2).

Proof of (5). The statement means that given a system $0 \to \mathcal{F}_i \to \mathcal{G}_i \to \mathcal{H}_i \to 0$ of exact sequences of abelian sheaves over a directed set I the sequence $0 \to \operatorname{colim} \mathcal{F}_i \to \operatorname{colim} \mathcal{G}_i \to \operatorname{colim} \mathcal{H}_i \to 0$ is exact as well. A formal argument using Homology, Lemma 05AA and the definition of colimits shows that the sequence $\operatorname{colim} \mathcal{F}_i \to \operatorname{colim} \mathcal{G}_i \to \operatorname{colim} \mathcal{H}_i \to 0$ is exact. Note that $\operatorname{colim} \mathcal{F}_i \to \operatorname{colim} \mathcal{G}_i$ is the sheafification of the map of presheaf colimits which is injective as each of the maps $\mathcal{F}_i \to \mathcal{G}_i$ is injective. Since sheafification is exact we conclude. □